Glass Heroes v0.3.4 — Inventory, Procurement & Stockroom Operations

Review status and identities

Development candidate for PR review. EXPERIMENTAL — GTA TESTING NOT RUN. This is not an RC package, certified stable build, crash fix or installation approval.

Branch: feature/shvdn3-340-inventory-procurement.

Audited RC1 parent: c8617aed5cda492214bbbe1907042bcafbfa0dc1 (owner RC1 runtime evaluation ongoing).

Source implementation commit: 1ded7ce2d0c35a7d9ddf9b0b652c37fbb7e53573.

Final test-adapter commit: 050725963961a5b8a0a3468372dbd64b94002870.

Source SHA256: fbc8f7f963bbffa3e31c097bf34250fef5807d28dd5de3b7d75463146beb3251.

Source: experimental/shvdn3/Install/scripts/GlassHeroesBusiness.3.cs.

Matching project: experimental/shvdn3/VisualStudio/GlassHeroesBusiness.V3.Experimental.csproj. It still compiles the install source directly; no duplicate script copy was introduced.

The owner first required YES/NO gates, approved Phase 0, then explicitly authorized work through all phases until ready for PR. That later authorization permits this development sequence. Merge, RC packaging, tags, public release, installation and owner-data modification remain outside this work.

Implemented behavior

A: inventory visibility

Inventory Overview retains its eight rows and existing keyboard/controller route. Category details show on-hand, repair-reserved, free and incoming quantities; existing estimated unit acquisition cost; and reorder advice. A compact four-category snapshot labels its columns. Shared storage reports placed-pallet unit usage, free units, incoming/projected headroom, free historical parcel slots and a capacity hold reason.

Reservations still come from UnreservedRepairStock, including physical jobs, remote jobs, autonomous workers and restored accepted promises with shared-quote deduplication. Incoming never becomes usable stock. Cost labels are estimates, not historical lot valuation or cost of goods sold.

Advice uses free stock plus incoming against the existing threshold, with the existing universal target at least five. It does not edit the cart or place an order. The existing Fill Low-stock Cart action keeps its original behavior; this milestone does not silently rewrite purchasing policy to match the new advice.

All projections are on demand. Long display totals use wide arithmetic and negative legacy counts are clamped only in the display; saved stock is not changed by browsing.

B: purchasing and order visibility

Build Supplier Order retains its four quantity controls and supplier cycling. Its comparison/preview action shows existing supplier lead times, requirements and current cart quote. Review Purchase Order opens a separate three-row page: details, confirm, Back.

Review snapshots quantities, supplier and quote without saving. Confirm requires unchanged supplier/quote and runs the existing capacity, one-pending-order and wholesale-vehicle checks. Back and failed placement retain the draft. Successful placement creates one pending mixed order, one ETA and an invoice only after stocking. No new prices, margins or expense formula.

Active Deliveries retains five rows. Its right column shows a short current state once ETA is ready. Requested notices include the full pending identity and explain holds. Order / Last Receipt shows current lines when pending, otherwise the durable last terminal record. Cancellation's second press is bound to the current order identity.

C: receipt durability and physical recovery

Every CommitSupplierDelivery call supplies an expected identity. Zero is a valid legacy identity and no longer bypasses comparison. Player/physical manager cargo uses its captured identity; synchronous virtual/direct receipt passes the current pending identity. Wholesale cannot finalize before its manual unload checkpoint.

Before mutation, the receipt validates order form, bounded quantities, frozen cost, mixed total, arithmetic, unit/parcel capacity and selected pallet. It applies all lines, supplier liability, earned provenance and terminal outcome, clears pending state and saves one complete snapshot. Save failure restores every changed accounting/order/outcome field. Notifications, earned-prop rebuild and departure follow durable success; retry cannot re-post a cleared order.

Driver exit issues one ordinary leave task after four active seconds, one fallback after 25, and holds/retries the original order after 40 if still blocked. Rear unloading and driver box creation require a valid on-foot driver. The box is not granted simply because 16 seconds elapsed before reaching the rear; the existing 45-second rear-stage recovery remains. No occupied vehicle is teleported by this change.

Wholesale load/unload checkpoints and manual recovery now check save success and roll back flags before finalizing visual ownership. Pallet purchase rolls back cash/spend/ownership on failure. Existing reload policy remains: unloaded/carried wholesale cargo is recovered at the warehouse, retaining the original order; no free receipt or automatic manager collection.

D: manager integration

Existing eligibility remains open shop + employed manager + placed shop, LOCAL/EXPRESS only. Nearby physical delivery and offsite virtual receipt share the same guarded commit. The existing distance predicate remains unchanged, so this milestone does not add a physical/virtual dead range. Player-carried cargo has priority and cannot be stolen by virtual receipt.

Manager pallet receipt rejects stale identity before native/pallet work. Live labels distinguish manager drop from player pickup. On-demand explanations cover ETA, player ownership, capacity, warehouse requirements, nearby physical configuration, driver exit/rear deadlines and streaming/save retry. No new manager-ped or wage gate; offsite simulation does not require an entity to stream.

E: integration and future readiness

Tests combine receipt liabilities with actual FinanceV1 observation and payments. Remote sales remain a subset of customer revenue, not an extra amount to add again. A $5,000 remote sales increase without outflows increases cash by $5,000. Invoice accrual at receipt does not debit cash. Actual bill payment is a separate outflow.

The v0.3.5 readiness document maps existing and proposed damage parts and describes future diagnosis, immutable requirements, reservations and idempotent repair application. No new item category, damage generation, physical restoration or speculative catalog is implemented.

Source map and before/after examples

Use method names in the pinned source; line numbers can shift across commits. Full unified diff is included as source.diff.

Methods / data	Change
PendingSupplierInventory, OnHandInventory, InventoryAvailabilityRow	Existing Phase A projection retained/refined.
InventoryStorageRows, InventoryReorderSuggestion	Shared unit/parcel forecast and advice, no writes.

Methods / data	Change
ShowInventoryCategoryDetail, ShowDetailedInventory, ShowReorderAdvice	Read-only notices and compact labels.
PrepareSupplierOrderReview, ShowSupplierPurchaseReview, ExecuteSupplierOrderReview	Frozen preview and separate explicit confirmation.
GetMenuItems/Value/Title, GetSelectedDescription, NavigateBack, ExecuteSelection	One new review page; existing row count/order otherwise retained.
SupplierLiveState, SupplierReceiptBlocker, MuleStageName	Current-state visibility and explanations, no new Tick monitor.
CommitSupplierDelivery, TryCancelSupplierOrder	Mandatory identity, checked posting, outcome in same transaction and rollback.
UpdateSupplierDriverExit, BeginSupplierRearUnload, GiveBoxToDriver	Bounded exit recovery and on-foot/rear validation.
TryCommitWholesaleCargo, TryResetSupplierCargo, TryBuyStoragePallet	Explicit checkpoint/purchase save-or-rollback.
SupplierOutcome; CaptureSupplierOutcome; LoadSupplierOutcome; WriteSupplierOutcome; ShowLastSupplierOutcome	Bounded optional last terminal record.
Save / constructor	One additive writer / pure metadata read plus new DEV label; existing payload/native startup sequence preserved.

Before: optional identity treated zero as “skip validation.”

```
CommitSupplierDelivery(int palletSlot = -1, long expectedOrderTicks = 0L)
// reject only when expectedOrderTicks != 0 and does not match
```

After:

```
CommitSupplierDelivery(int palletSlot, long expectedOrderTicks)
// reject whenever expectedOrderTicks != _deliveryDueUtcTicks
```

Before: blocked exit submitted fallback then immediately began rear unloading. After: submit the fallback once, verify on-foot state on a subsequent active step, otherwise defer at the deadline. Rear proximity, not elapsed time alone, authorizes box creation.

Before: cargo flags / pallet purchase called Save without checking its result. After: helpers snapshot changed fields, save once, restore them on false, then let the caller finalize visual state.

Receipt and finance state model

Cart -> Review -> durable Pending -> active ETA -> physical / eligible virtual / player wholesale cargo -> validated receipt -> one atomic save -> Received -> visual departure.

Blocked/missing entities -> held original order -> bounded retry. Failed receipt save -> original pending order and prior outcome restored. Cancel -> durable pending-clear + optional prepaid refund + last cancellation. Paid invoice -> separate actual cash outflow.

The saved receipt is authoritative for finalization; a truck, crate, menu notification or best-effort CSV is not evidence to post stock again. Supplier order IDs and FinanceV1 transaction IDs remain separate namespaces; no guessed foreign key is written.

Persistence and backward compatibility

Only additive SupplierOutcomeV1 keys are introduced. Version 1 stores State (0 absent, 1 received, 2 cancelled), OrderTicks, Cost, Invoice, ActiveMs, Supplier, Pallet, Prepaid and Line0-Line3. Pallet defaults to -1 for no physical slot. Absent or invalid optional history means unknown history, not fabricated receipt. It does not invalidate a valid pending legacy order.

History stores one last terminal outcome, not an unbounded order archive. ActiveMs comes from the existing active-game clock; OrderTicks remains opaque identity and never drives ETA. Existing SupplierOrderV1, legacy Purchasing fields, FinanceV1, RepairPromisesV1, stock, settings, wages/progression and placement keys keep

their meanings. No settings key is added.

The complete Save snapshot differs only by WriteSupplierOutcome; the established temporary-write/atomic-replacement helper is unchanged. Full source preservation checks enforce this and protect original v2 source/projects, finance/quote types and 361 unchanged methods.

A receipt accrues an unpaid supplier liability unless prepaid; it is not a cash payment. Prepaid cancellation refunds once; unpaid cancellation has no refund. New history does not invent previously completed order identities or acquisition lots.

Bounds, clocks and crash safety

Keep one pending mixed purchase order, four categories, four pallets, nine historical parcel slots per pallet, one supplier truck/driver/in-flight box and existing model/visual retry bounds. No new models, worker threads, frame scans or per-Tick logging.

OnTick, active/unpaused time sampling, manager income, demand, wages and saved remaining ETA are unchanged. Offline and paused progress remain zero. Constructor changes are metadata reading and startup label only, with source SHA256 identification and gameplay gate preserved.

Clock and persistence assertions are automated evidence, not actual GTA execution. CLR stack overflow, native startup access violation and Read-only/EUP behavior remain separate investigations and UNATTRIBUTED. No crash fix or new stability claim.

Actual verification

Release: net48/C# 7.3 against official SHVDN3 3.6.0, zero errors, inherited warnings CS0169 and CS0649.

862/862 behavior checks, including 184 production navigation checks. 21/21 static checks and 13/13 inventory structural checks, counted separately. New behavior suites: inventory 30, procurement review 12, receipt/checkpoints 32, manager 13, finance integration 9. Existing substantive assertions retained; legacy adapters gained the new mandatory receipt dependencies/identity parameter.

The original baseline had 762 behavior checks. The new total adds 96 new suite cases plus four navigation cases (new page and three targeted input scenarios). Adapter tests execute extracted production methods with deterministic world/file adapters under .NET 8; the full script separately compiles under net48/C#7.3. They do not execute GTA pathfinding, native attachment, actual game-process interruption or the installed Enhanced runtime.

Run in the managed workspace:

```
source /workspace/.cloud-setup/env.sh
python verification/inventory340/validate.py --output /tmp/gh340-review-validation
```

The environment activation is specific to this workspace; other machines need the .NET SDK, net48 reference assemblies and pinned NuGet dependency. Commands, logs, source hashes and preservation results accompany this guide.

Known limitations and deferred work

- Owner RC1 runtime evaluation and v0.3.4 GTA acceptance remain pending.

- One pending order and one last outcome; no multi-order queue/history archive.

- Supplier cycling retained; no new full supplier catalog screen or dynamic quote engine.

- Nearby manager auto-drop still requires configured physical Mule route; no new per-order attended override.

- Earned boxes remain after sales/consumption, so parcel slots can block receipt with unit headroom. No silent visual removal/rotation.

- Wholesale carry reload uses existing warehouse recovery, so the owner may need to repeat transport; it does not recreate inventory or invoices.

- Notification wrapping at actual resolution/UI scale and physical route recovery need GTA confirmation.

- No new damage categories, COGS/lot valuation, retail markup, automatic buying, customer delivery missions, manager wholesale, tablet or physical damaged-car repairs.

Review, future testing and rollback

Review against feature/shvdn3-336-service-recovery-business-polish, not main. Keep the working RC1 and prior v0.3.3.5 baseline available. No merge or release is performed.

When an experimental test package is separately approved: close GTA; back up the matching script and the entire scripts/GlassHeroesBusiness directory, including Save.ini, Settings.ini, FinanceV1 data in Save.ini and every companion CSV/journal/profile file present. Keep backups outside scripts. Install exactly one GlassHeroesBusiness.3.cs; move other Glass Heroes .2.cs/.3.cs/DLL versions out of all script search locations. Do not install reference/dependency DLLs just because the project compiled with them.

Use a disposable complete data copy and the owner Legacy 1.0.3889.0 / Enhanced 3.9.0.6 / LSRED setup. Follow GTA_TEST_CHECKLIST.md and record installed hash/startup identity. Do not alternate experimental/baseline scripts against the same modified sole Save.ini.

Rollback: close GTA, archive the complete experimental data for evidence, remove its script, restore the matching full baseline business directory and baseline script, verify hashes and exactly one script, then launch. Unknown-key preservation does not prove old builds understand new receipt metadata; restoring only an old Save.ini is insufficient.

v0.3.4 development changelog — experimental, GTA untested

- Inventory details show on-hand, reserved, free, incoming, estimated cost and non-purchasing reorder advice.

- Shared unit capacity and historical parcel slots shown separately; earned boxes preserved.

- Added explicit mixed supplier purchase review/confirmation with frozen quote/supplier validation and Back retaining draft.

- Added compact live order state, full pending identity and durable last receipt/cancellation.

- Receipt requires captured identity, validates all mixed quantities/capacity/arithmetic, and saves outcome with stock/liability/pending-clear atomically.

- Driver exit is confirmed before rear unload; retries bounded; rear proximity required before driver box creation.

- Wholesale load/unload/recovery and pallet purchases roll back failed saves.

- Existing manager LOCAL/EXPRESS nearby/offsite receipt integrated with shared guarded commit; wholesale remains player-only.

- Added 96 behavior cases and four navigation cases; full total 862 passing, plus 21 static and 13 inventory structural checks.

- Documented v0.3.5 damage-parts relationships only; no new items or physical restoration.

- Existing prices, financial engine, repair reservations/settlement, placements, employee progression and active-unpaused clocks preserved.

No merge, tag, release package, public release or live installation. Crash investigations remain separate and unresolved.

v0.3.4 GTA acceptance checklist — NOT RUN

For a future separately approved experimental test package. Do not install from these instructions without that approval. No RC prepared by this development task.

Controlled setup and backup

- [ ] Close GTA. Copy the installed script and complete scripts/GlassHeroesBusiness directory to a dated backup outside scripts. Include Save.ini, Settings.ini and all companion files; FinanceV1 is embedded in Save.ini.

- [ ] Preserve working v0.3.3.6 RC1 and v0.3.3.5 rollback artifacts. Use a disposable full data copy; never alternate builds against the same modified sole save.

- [ ] Record GTA Legacy 1.0.3889.0, SHVDN Enhanced 3.9.0.6, LSRED/mod versions/settings and backup identity. Keep other mods constant for comparison.

- [ ] Verify installed source SHA256 fbc8f7f963bbffa3e31c097bf34250fef5807d28dd5de3b7d75463146beb3251 and startup DEV identity. Install exactly one Glass Heroes script, with older versions/DLLs outside script loading paths. No extra reference/runtime files.

- [ ] Cold launch and confirm existing gameplay startup gate, no script reload/initialization/save errors. Net48 compilation does not certify Enhanced runtime compatibility.

Inventory and input

- [ ] F2 > Business Management > Inventory & Suppliers > Inventory Overview: enter each category; compare on-hand/reserved/free/incoming against a copied save and accepted physical/remote/autonomous jobs.

- [ ] Verify reserved universal fallback is counted once; incoming cargo cannot satisfy repair or retail availability.

- [ ] Inspect cost estimates, threshold and reorder advice. Browsing changes no cash, stock, pending identity or liabilities and places no order.

- [ ] Storage summary shows unit use/free/projected headroom and parcel slots. Nine small historical deliveries can fill slots even with units free; sales/repairs do not erase earned boxes.

- [ ] Read notices at actual UI scale; report wrapping/clipping with source identity. Check every changed row footer.

- [ ] Keyboard and controller quantity controls, wrap, review, confirm and Back. Back from supplier review restores cart row 6 without purchase. Esc/Start hides overlay and opens native pause without transaction.

Purchase confirmation and accounting

- [ ] Select each supplier; compare configured ETA, current quote and requirements. Quantities/price formulas remain unchanged.

- [ ] Build a four-line order, open review, Back/edit/review, then confirm once. One pending identity, one ETA, no receipt stock or unpaid-invoice cash debit at placement.

- [ ] Held Enter/Accept cannot create duplicate orders; second pending order is rejected.

- [ ] Cancel unpaid order: no stock/invoice/refund. For an existing legacy prepaid fixture, cancel refunds once. Never fabricate a prepaid live order by editing sole data.

- [ ] If another order replaces an old completed/cancelled one, cancellation requires confirmation for the new identity.

- [ ] Verify receipt increments every mixed bucket once, one parcel provenance slot and one supplier liability unless prepaid. Reload and inspect Order / Last Receipt and current pending identity separately.

- [ ] Last receipt says invoice accrued, not paid; actual payment under Finances & Accounting reduces cash/liability once. Compare opening + inflows - outflows = closing.

- [ ] Observe remote sales over a controlled shift: with $5,000 new remote revenue and no outflows expect +$5,000 cash; with payments expect the corresponding net increase. Remote revenue must not be added twice to customer totals.

Physical and manager delivery

- [ ] LOCAL then EXPRESS with manager absent: arrival, parking, confirmed driver exit, rear approach/unload, attached driver box, handoff, player carry, saved pallet deposit, receipt notification and departure.

- [ ] Block driver exit safely: one fallback attempt, then recovery after 40 active seconds. Order stays pending, no stock/invoice posted.

- [ ] Block rear access: no early box solely from elapsed time; recovery at 45 active seconds. Missing driver/vehicle/crate/model leaves original order recoverable.

- [ ] Hire manager/open shop, stay nearby: driver follows saved stops and deposits automatically at valid pallet. Status distinguishes manager drop from player pickup.

- [ ] Move across the near/away boundary both directions during arrival/walking; no dead range or duplicate receipt. Player-carried package cannot be auto-received offsite.

- [ ] Away, manager hired/open: virtual LOCAL/EXPRESS receipt after active ETA with capacity. Closed/no manager never receives through manager simulation.

- [ ] Nearby missing physical route, full pallet/unit limit, no parcel slot or absent pallet support: clear hold, no duplicate invoice and recoverable order.

Wholesale, persistence and regression

- [ ] WHOLESALE requires assigned vehicle; player warehouse pickup, load, transport, unload, deposit. Manager never collects or stocks warehouse cargo remotely.

- [ ] Reload at ETA, warehouse carry, vehicle-loaded and player-unloaded checkpoints. Existing carried/unloaded reload recovery may require collecting at warehouse again; original order/cost stays, no receipt until stocking.

- [ ] Test failed-save behavior only in a controlled disposable environment; do not alter live permissions/attributes. Failed cargo/pallet/receipt actions retain prior cash/stock/liability/order and remain retryable.

- [ ] Reload after successful receipt: outcome/stock/invoice remain, old order cannot receive again. Missing cosmetic prop never authorizes another credit.

- [ ] Pause longer than ETA; close GTA for a long interval and reload. No paused/offline income, wages, delivery or virtual progress. GTA clock changes alone cannot create catch-up.

- [ ] Run 1/2/3 mechanic queues, accepted repair quotes, physical transfers ON/teleport mode OFF, wages, owner transfers, parts retail and all saved role paths/points/headings unchanged.

- [ ] Extended Glass Heroes + LSRED session: bounded entities/retries, no growing orphaned trucks/boxes, performance comparable to baseline. Record exact uptime and activity.

- [ ] If crash occurs, preserve matching dump/analyzer/SHVDN/RPH/Glass Heroes logs and installed hash. Keep CLR overflow, startup AV and Read-only/EUP symptoms separate; do not attribute based on last logged action.

- [ ] Prove rollback with GTA closed: archive full candidate data, restore matching complete baseline directory + script, verify exactly one script and baseline hash.

v0.3.5 damaged-car inventory readiness — design only

v0.3.4 does not implement physical damage generation/restoration, new parts categories, dynamic pricing or a new repair settlement engine.

Current contracts

The four stored buckets remain Universal Parts, Service Filters, Brake Kits and Diagnostic Supplies. RepairQuote owns ServiceKind, DedicatedUnits, UniversalUnits, OrderId, agreed estimates/total and billable duration. UnreservedRepairStock derives availability across physical/remote/autonomous/restored accepted work, deduplicating shared quotes. TrySettleRepair consumes supplies and posts cash/progression atomically; ProcessOneCustomerRepair requires the legitimate mechanic/bay/completed-work state. Current service names are Routine Service, Brake Service and Engine Diagnostics. These contracts are preserved verbatim by preservation.py.

SupplierOrderV1 remains four-line/one-pending; SupplierOutcomeV1 is a bounded terminal procurement audit, not a repair-plan or stock-lot schema. Incoming cannot be consumed or sold. Pallet visuals record earned

receipts rather than current units; quantity sales do not delete boxes.

Proposed future mappings — owner approval required

Damage domain	Existing item relationship	Additional design/schema if approved
Bodywork / panels	No dedicated item; universal is not a panel counter	Stable item IDs, panel/kit unit semantics, supplier/cart lines, reserved quantities and save version.
Engine components	Diagnostic supplies are diagnostic consumables, not replacement engine components	Separate component definitions and diagnosis recipes; do not reinterpret existing stock.
Tires / wheels	No dedicated item	Decide tire vs wheel granularity, quantity, supply and persistence.
Brakes	Existing Brake Kits may fit	Approve kit semantics and component substitution before reuse.
Glass / windows	No dedicated item	Decide kit vs individual window, supported restoration and stock quantities.
General maintenance	Universal Parts / Service Filters	Reuse current meaning/pricing; freeze accepted requirements.

Do not append categories to existing four-element arrays without auditing every cart, receipt, quote, reservation, retail, capacity and migration consumer. v0.3.4 makes no speculative catalog changes.

Future relationships and sequencing

Customer order identity -> bounded versioned DamageAssessment -> accepted RepairPlan with immutable PartRequirement lines -> job-owned reservation -> independent mechanic/bay assignment -> active-time work -> durable settlement/repair authorization -> idempotent authorized restoration -> return/pickup.

The vehicle must remain damaged through collection and initial bay parking. Do not repair it on spawn, transfer, reservation or animation start. Parts must be reserved/available for the accepted plan. Final restoration occurs only after successful service.

Disk and GTA natives cannot be committed atomically. Proposed future policy: durably settle parts/finance plus repair authorization once, then apply bounded component-specific restoration; hold return and retry if natives fail, without consuming or paying twice. If the durable commit fails, preserve damage and restore accounting. This ordering, failure compensation, lost-vehicle handling and supported damage operations require separate v0.3.5 approval.

Persist order/plan identities, never entity handles. Diagnose only owned customer vehicles at transitions, not every Tick. Broad fix-everything natives may erase unrelated damage and must not replace an approved component plan. Legacy accepted quotes retain their contract; no retroactive new requirements or repricing.

Future acceptance gates

1. Owner approves exact new items, substitutions, supplier/retail policy and storage semantics.

2. Versioned repair-plan migration defaults legacy jobs to their existing promises.

3. Damaged car stays damaged until work completion in both transfer modes.

4. Reservations prevent retail from consuming planned parts.

5. Failed save leaves stock/cash/damage unchanged; native failure after durable settlement holds return and never repeats payment/consumption.

6. Missing entities, reload, pause and long closed-game intervals do not fabricate work, parts, vehicle restoration or revenue.

7. Legacy/Enhanced/LSRED engine compatibility and actual damage APIs are validated in GTA.